



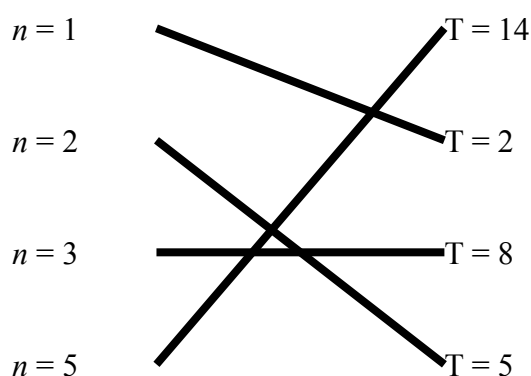
Expressing an arithmetic sequence

Task A: The position of a term in a sequence

1) Match the term number (n) to the term values (T). One has been done for you.

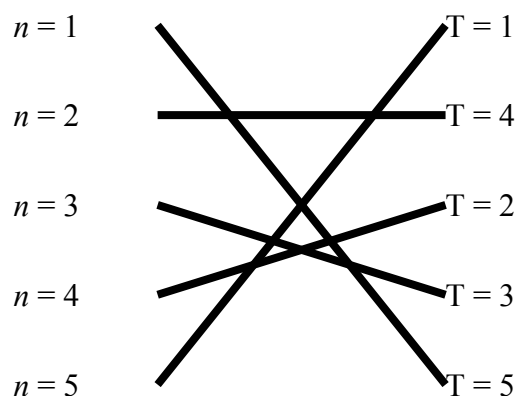
a)

Term number (n)	1	2	3	4	5
Term value (T)	2	5	8	11	14



b)

Term number (n)	1	2	3	4	5
Term value (T)	5	4	3	2	1



2) Andeep and Jacob are looking at an arithmetic sequence.

Term number (n)	1	2	3	4	5
Term value (T)	45	40	35	30	25

I see the pattern!
When $n = 0$ the term is 50

I agree! Next would be
when $n = -1$
The term will be 55



Write a sentence explaining why Andeep and Jacob are wrong.

You might have written:

There is no zero term, or -1 term in this sequence.

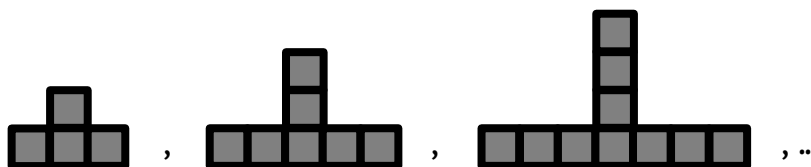
The first term 45, where $n = 1$, is the first number in this sequence.

There is nothing before it.



Task B: The general term of a sequence

1) Sam spots a general rule for this sequence.



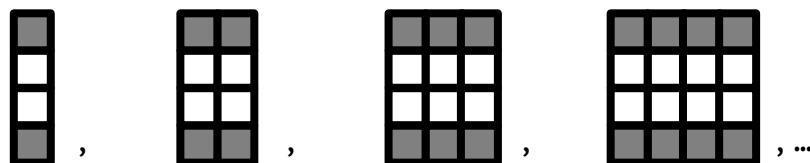
Pattern number	1	2	3	Pattern number
Squares	4	7	10	Pattern number $\times 3 + 1$

Whilst their general rule works, how would you improve upon it?

You might have said, use n to represent the n^{th} term thus writing the rule more concisely.

Pattern number (n)	1	2	3	n
Squares	4	7	10	$n \times 3 + 1$

2) Fill in the table for this sequence and write a general rule each row.



Pattern number (n)	1	2	3	4	5	25	50	100	n
Shaded squares	2	4	6	8	10	50	100	200	$n \times 2$
Total squares	4	8	12	16	20	100	200	400	$n \times 4$
Perimeter	10	12	14	16	18	58	108	208	$n \times 2 + 8$